VIDEL-GS : Visual-Inertial Drone Environment Localization and Gaussian Splatting

Sponsor : Student self-direct, no sponsor

Background:
Whether it's surveying a space, virtual tours, or video game assets, getting a 3d scene from a real-world environment has its uses. With recent advances in sensor fusion and drone technology, our goal is to make mapping rooms and environments easy and intuitive.

Project Needs:
1. Remote-controlled drone, easy to modify and is financially accessible.
2. A board attachment for the drone with the necessary sensors: camera(s) and imu.
3. A small form factor machine that will receive the sensor data.
4. Software that will merge the data and generate a gaussian splat of the environment.

Project scope:
1. On-the-drone software that would transmit sensor data to the target machine.
2. Software for any machine a user would like to use to capture and process the sensor data.
3. That same software being easy to use and makes use of all hardware capabilities.
4. Documentation to make this project reproducible.

Potential Technical Considerations:
1. Refactoring source code from papers for efficiency/deployment (aka any conversions to C++ and for models Onnx/TensorRt).
2. Transmitting sensor data across Bluetooth or Wi-Fi if on the same network, if not possible then potential saving data for offline processing.
3. The assembling of not only the board, but also connecting it to the drone, will take some engineering skill.

Deliverables:
1. Software prototypes
2. On-the-drone board prototypes
3. Final prototype, combining both of the previous

Assumptions and Dependencies:
TBD